Master Giving Acknowledge to a Slave Device

- Upon reception of a byte from a slave, the master must acknowledge this to the slave device.

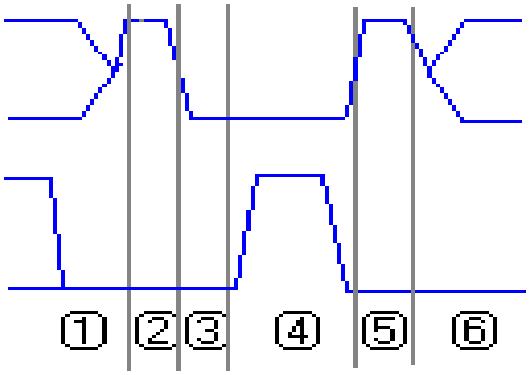
  The master is in full control of the SDA and the SCL line.

  After transmission of the last bit to the master (1) the slave will release the SDA line.

  The SDA line should then go high (2). The Master will now pull the SDA line low (3) .

  Next, the master will put a clock pulse on the SCL line (4). After completion of this clock pulse, the master will again release the SDA line (5).

  The slave will now regain control of the SDA line (6).



```
u8 i2c_masterack(void)
{
     //Assert Ack
    I2CONSET = 1<<AA_BIT;
    //Clear SI  bit
    I2CONCLR = 1<<SI_BIT;
    //Wait for SI Bit
  while(((I2CONSET>>SI_BIT)&1)==0);
    //Clear Assert Ack
    I2CONCLR = 1<<AA_BIT;;
    //read read byte
    return I2DAT;
}
```

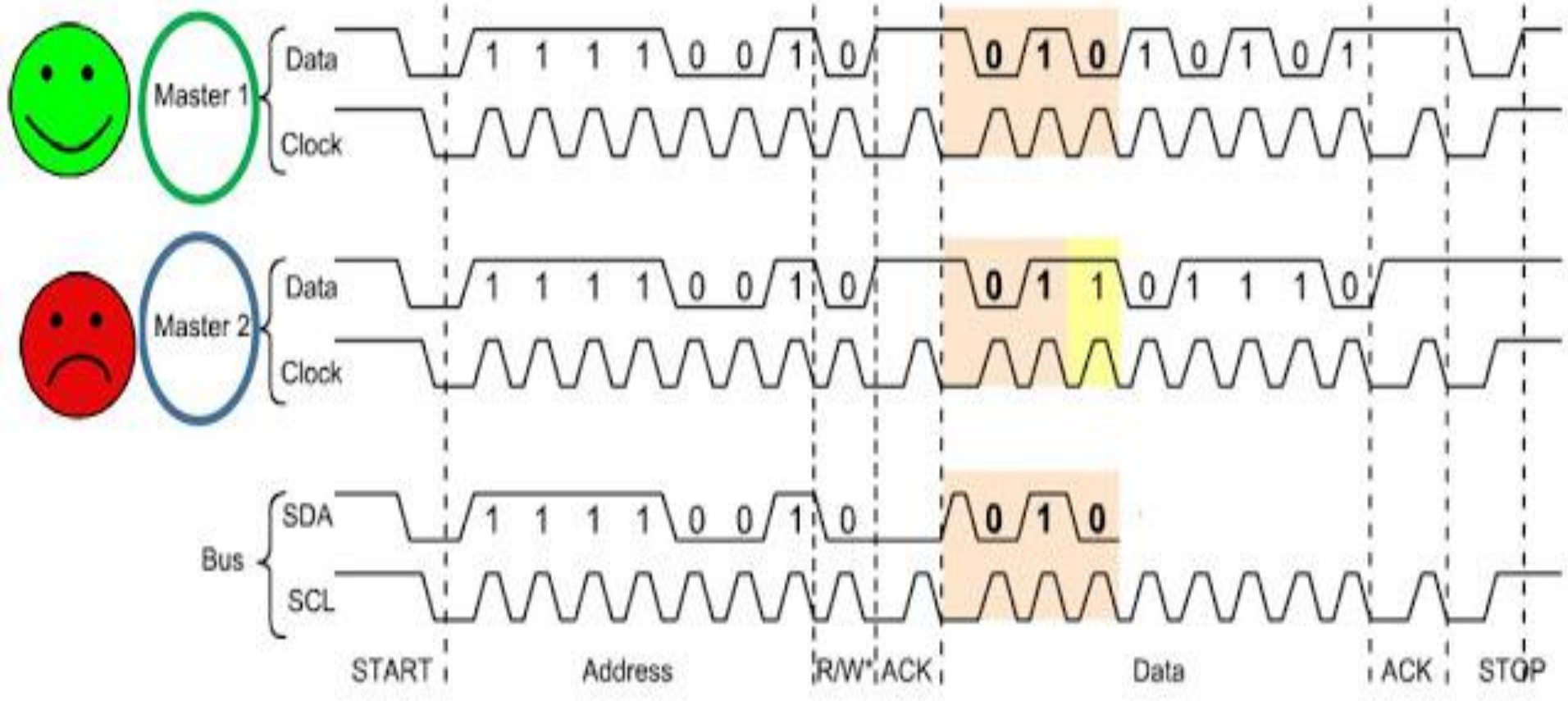
```
void i2cMAck(void)
{
    SCLRBIT(IOCLR0,SCL);
    SCLRBIT(IOCLR0,SDA);
    SSETBIT(IOSET0,SCL);
    SCLRBIT(IOCLR0,SCL);
    SSETBIT(IOSET0,SDA);
}
```

```
u8 i2c_masterack(void)
{
     //Assert Ack
    I2CONSET = 1<<AA_BIT;
    //Clear SI  bit
    I2CONCLR = 1<<SI_BIT;
    //Wait for SI Bit
    while(((I2CONSET>>SI_BIT)&1)==0);
    //Clear Assert Ack
    I2CONCLR = 1<<AA_BIT;;
    //read read byte
    return I2DAT;
}
```

I2C Bus Arbitration

- Refers to scenario When 2 or more I2C Masters operate concurrently to use the bus, which master eventually wins the control over the bus to complete its transaction without corruption of data.
- I2C bus arbitration is achieved due to wired-and-logic.
- Logic 0 is dominant,
- Logic 1 is recessive.
- During the transfer, the masters constantly monitor SDA and SCL. If one of them detects that SDA is low when it should actually be high, it assumes that another master is active and immediately stops its transfer.

Bus Arbitration



Clock stretching

- Is a mechanism used by an I2C slave to notify the master to slow down the clock speed.
- *I2C devices can slow down communication by stretching SCL: During an SCL low phase, any I2C device on the bus may additionally hold down SCL to prevent it from rising again, enabling it to slow down the SCL clock rate or to stop I2C communication for a while.*
- *This is also referred to as clock synchronization*.

Need for Clock Stretching

- This situation may happen when the slave is waiting for its internal operation to finish, such as EEPROM read/write, sensor data calculation, internal state machine to settle down, and so on.
- *The master, on the other hand, is required to read back the clock signal after releasing it to the high state and wait until the line has actually gone high.*

I2C Clock Generation and I2C Clock Synchronization (cont.)

- If a device pulls down the clock line for a longer time
  - All clock generators must enter the WAIT-state
    - A slave can slow down a fast master
      - The slow device can create enough time to store a received byte or to prepare a byte to be transmitted (Clock Stretching)

